

Data Plots: Hydrolysis of Nitrosamines

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This exploration provides discussion for one of the cases presented in class meeting #14. In the class handout you were asked make a Hammett plot and come to a conclusion about the reaction mechanism. This document explores that data and ways to interpret it by plotting and fitting the data to mathematical models.¹

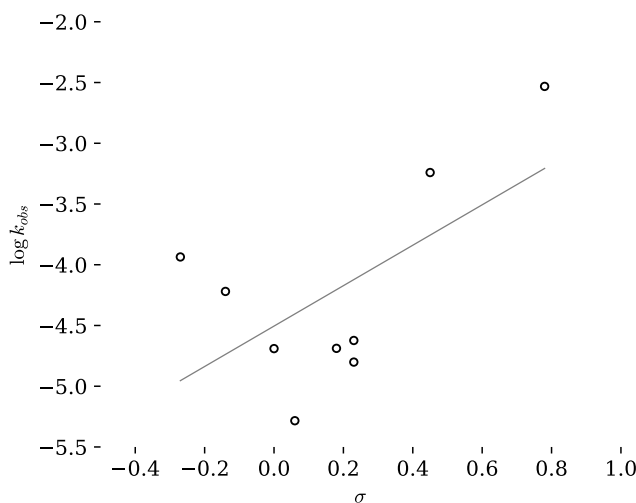
Plots: Curved Hammett Plots

This data comes from an investigation into the mechanism of hydrolysis of nitrosamines^{2,3} I found this example data in a textbook but was unable to get access to the original publication in a short-lived Latvian chemistry journal from before the fall of the Soviet Union. I'm old, but I'm not that old.

This document will simply present the plots that were discussed in class. The plots are generated using a *Python* notebook available via Google Colab.⁴

The Basic Plot

Below is a Hammett plot of $\log k_{obs}$ vs σ . As you can see, a simple line plot may not be useful here.



This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

¹ Read chapter 8.3 and 8.4.

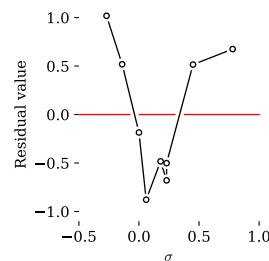
² "Physical Organic Chemistry, 2nd ed." by Neil Isaacs. 1995, Wiley, New York (the 1st edition is available in UPEI library stacks QD476.I846 1987.)

³ B.A. Porai-Koshits, E.Y. Belyaev, E. Shadovskii, *React. Sposobnost. Org. Soedin.*, 1964, 1, 10.

⁴ All plots in this exploration were generated using the interactive *Python* notebook available via Google colab at <https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.16/16.Class.Handout.Problems.Nitrosamines.ipynb>.

Figure 1: plot of $\log k_{obs}$ vs. σ . Do you see a bend in the plot? If we fit to a straight line we see that $\rho = 1.43 \pm 0.83$, $R^2 = 0.37$.

Figure 2: A plot of the residuals vs. σ for the line fit in figure 1. We observe a systematic variation from the model, not random noise.



Two Separate Plots

Below is a Hammett plot of $\log k_{obs}$ vs σ . As you can see, I divided the points into two groups and performed a line fit on each. This gave two separate Hammett equations.

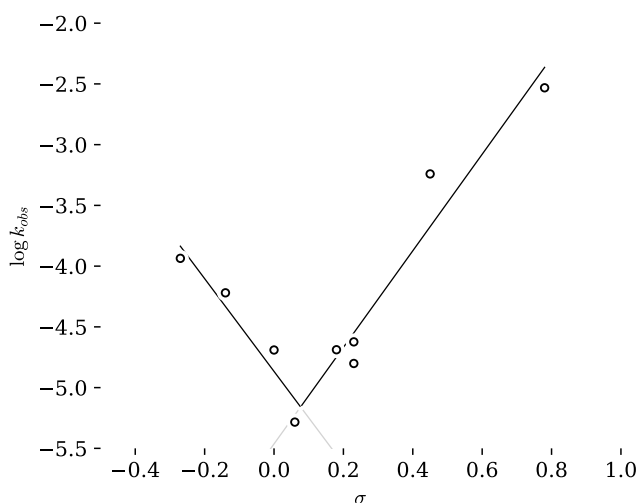


Figure 3: plot of $\log k_{obs}$ vs. σ . I bet you see the bend in the plot now. Of course, I am directing your eye by drawing the two lines.

$$y = (-3.82 \pm 0.83)x + (-4.87 \pm 0.13), R^2 = 0.914$$

$$y = (3.97 \pm 0.62)x + (-5.46 \pm 0.27), R^2 = 0.932$$

Adding up the Two Separate Plots

Below is a Hammett plot of $\log k_{obs}$ vs σ . The line fit is composed of the sums of the two Hammett equations. It made for a good fit overall.

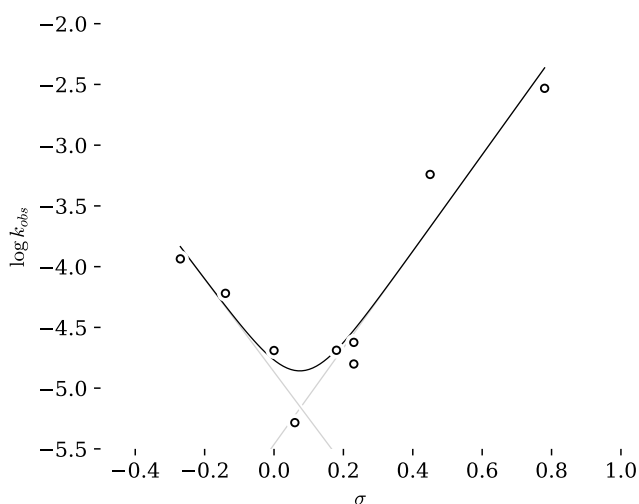
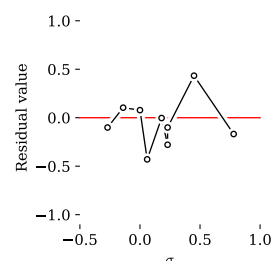


Figure 4: plot of $\log k_{obs}$ vs. σ . Here the two Hammett equations determined in figure 3 are summed. The fit looks very good. The R^2 value is 0.914 overall.

Figure 5: A plot of the residuals vs. σ for the line fit in figure 4. This is a better fit.



A Single Curve Fit

Below is a Hammett plot of $\log k_{obs}$ vs σ . The line fit is a model that is the sum of two Hammett equations. The curve fit will optimize for the pair of slopes and the pair of intercepts.

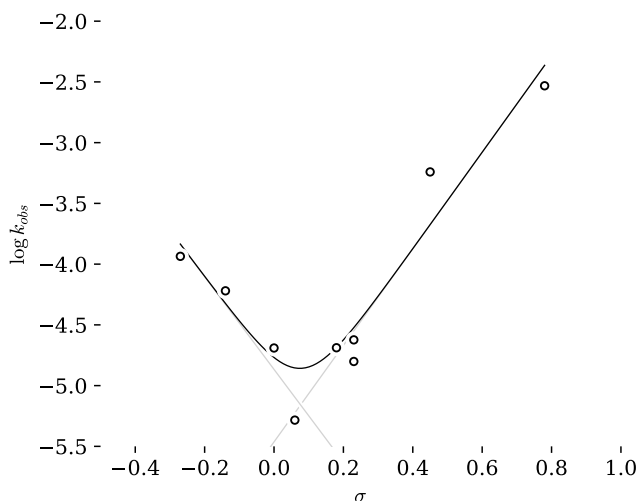
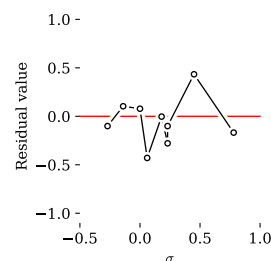


Figure 6: plot of $\log k_{obs}$ vs. σ . Here two Hammett equations are combined to make a single math model which was used for the curve fit. The fit looks the same as before, but it is very slightly different. Observe the large differences in standard deviation reported by the curve fit. The R^2 value is 0.921 overall.

$$\rho_1 = -4.26 \pm 2.67$$

$$\rho_2 = 4.15 \pm 0.42$$

Figure 7: A plot of the residuals vs. σ for the line fit in figure 4. This is just a good a fit as previously.



Observe the large difference in precision, especially for the ρ value reported for region with the more electron-donating substituents. The precision of the earlier pair of line fits was not really true. For the electron-donating series, all four points were close to the line. But the 4th point was well off the curve when the two rates were added. The curve fit uses that 4th point and it results in the “valley” of the plot being pulled downward. This pulls the line a bit away from the slope established by the three left-most points. This discrepancy results in the large standard deviation that was calculated. I wish I had paid attention during my statistics course when I was your age.

Interpretation

So how do you interpret a Hammett plot where the line is “concave up” and moves from a slope of about $\rho = -4$ to a slope of about $\rho = 4$? Answer the questions in the class meeting handout and you will find your way.